Problem 1a) (15 points)

Given the image region as shown in Figure 1 and $V = \{1\}$, what is the shortest m-path between p (the pixel at the upper-left corner) and q (the pixel at the bottom-right corner)? Give the length of the path if the path exists.

$$\begin{matrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 1 \end{matrix}$$

Step 1: Identify pixels in $V = \{1\}$

p

1	0	1	1	0
0	1	0	1	0
0	0	1	1	0
1	1	0	1	0
0	1	1	1	1

q

All pixels with value 1 are highlighted. These are the only candidates for the m-path since $V = \{1\}$.

Step 2: Apply m-adjacency rules

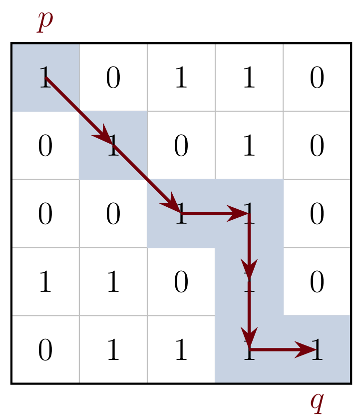
Two pixels with values in V are **m-adjacent** if:

a) They are 4-adjacent (share an edge), OR

b) They are diagonally adjacent AND their shared 4-neighbors contain no pixels with values in V.

The second condition prevents diagonal shortcuts when a 4-connected bridge exists, eliminating path ambiguity.

Step 3: Trace the shortest m-path



Path: $(0,0) \rightarrow (1,1) \rightarrow (2,2) \rightarrow (2,3) \rightarrow (3,3) \rightarrow (4,3) \rightarrow (4,4)$

Diagonal move justifications:

- $(0,0) \rightarrow (1,1)$: Shared 4-neighbors are $(0,1) = 0$ and $(1,0) = 0$. Neither is in V, so m-adjacent.
- $(1,1) \rightarrow (2,2)$: Shared 4-neighbors are $(1,2) = 0$ and $(2,1) = 0$. Neither is in V, so m-adjacent.

Results

The shortest m-path exists and has **length 6**.

Problem 1b) (15 points)

To take a picture for fireworks at night, give at least two operations that should be applied to improve the visualization quality of the image and explain.

Step 1: Understand the Scene Characteristics

Fireworks at night present specific challenges:

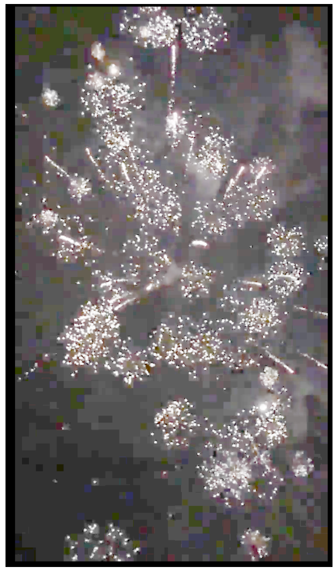
- **High dynamic range**: Very bright fireworks against a very dark sky
- **Desired appearance**: Pure black sky with vivid, crisp firework sparks
- **No hidden information**: The dark sky contains nothing meaningful to reveal

Original fireworks image

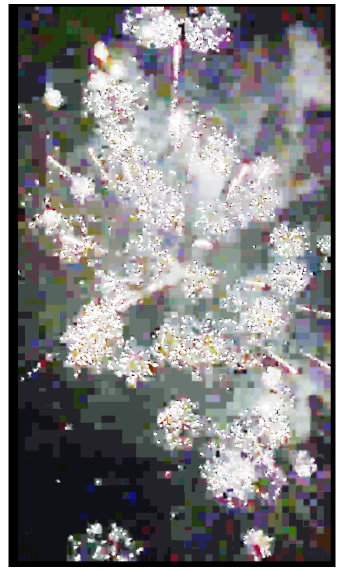
Step 2: Brightening Operations

These operations attempt to reveal detail in shadows by lifting dark intensities.

Gamma ($\gamma = 0.5$)



Log Transform

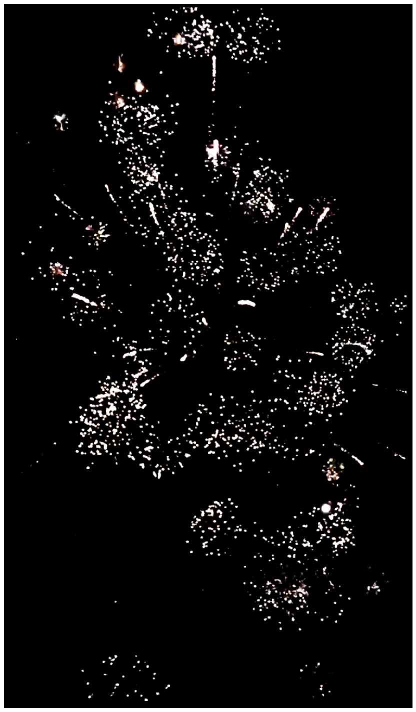


Histogram Equalization

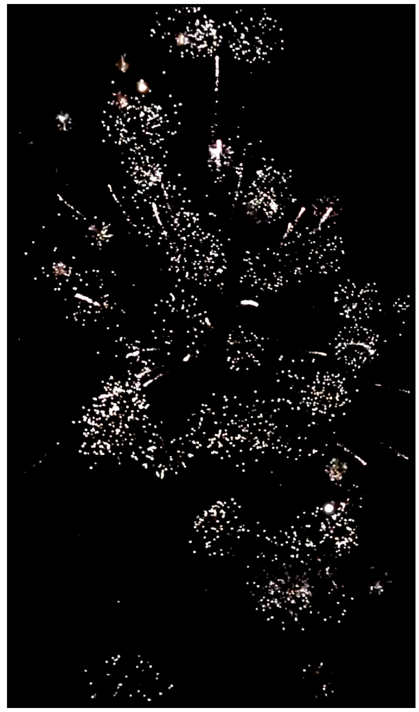
Effect: The black sky becomes gray, revealing noise and sensor artifacts. This is **undesirable** for fireworks because the sky *should* be black—there is no meaningful information to reveal.

Step 3: Contrast Enhancement Operations

These operations push dark values toward black and light values toward white.



S-Curve Contrast



Boosted Contrast

Effect: The sky becomes pure black while the fireworks remain bright. This is **desirable** for aesthetic night photography.

Step 4: Sharpening Operations

These operations enhance edges and fine details.

Unsharp Masking

High-Boost Filtering

Effect: Firework sparks appear crisper with more defined edges.

Step 5: Color Enhancement Operations


Saturation Boost


Warm Temperature


Warm + Color Contrast

Saturation: Uniformly boosts all color intensity—can cause clipping in already-vivid areas.
Warm Temperature: Shifts colors toward orange/yellow by boosting red and reducing blue channels, enhancing the fiery appearance.
Warm + Color Contrast: Applies warmth first, then an S-curve in Lab color space to push warm colors warmer and cool colors cooler, creating rich golden tones.

Step 6: Consider the Optimization Goal

The "correct" operations depend entirely on what you are trying to achieve:

Goal	**Recommended Operations**	**Avoid**
Aesthetic (fireworks)	S-curve, sharpening, saturation	Gamma, Hist EQ
Surveillance	Gamma, log transform, Hist EQ	S-curve
Medical imaging	CLAHE, careful contrast	Extreme transforms
Object detection	Thresholding, edge detection	Aesthetic enhancements

The same operation can be *beneficial* or *harmful* depending on the goal. Histogram equalization is excellent for revealing hidden details in surveillance footage, but terrible for fireworks photography where the dark background should remain dark.

Step 7: Consider the Display Medium

The target display device also affects which operations are most beneficial:

Display	Prioritize	Less Important
Mobile phone (small screen)	Color, contrast, brightness	Fine sharpening
Desktop monitor	Sharpness, fine detail	Can be subtle
Print	Resolution, color accuracy	Different gamma curve

On a mobile phone, viewers cannot resolve fine detail, so aggressive sharpening provides little benefit—but punchy colors and contrast help images "pop" on a small screen viewed in variable lighting. On a desktop monitor, fine details are visible, making sharpening more valuable.

Results

Recommended operations for fireworks photography:

a) **S-curve contrast enhancement**: Apply a sigmoid transformation $s = \frac{1}{1+e^{-k(r-0.5)}}$ to push dark pixels darker and bright pixels brighter. This ensures a pure black sky while maintaining bright firework sparks.

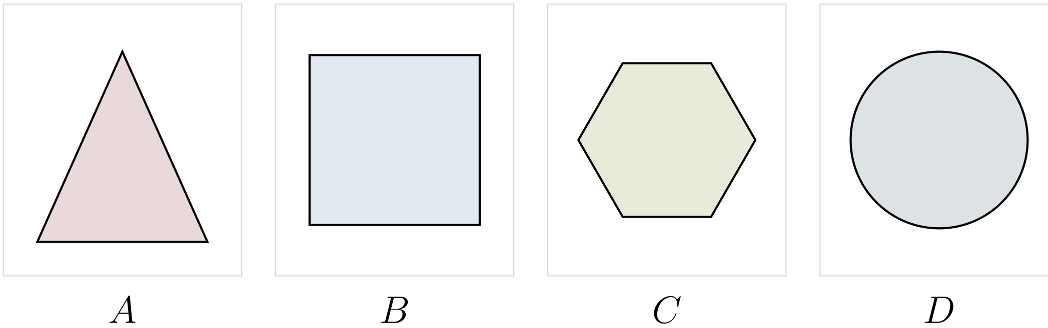
b) **Saturation boost** (especially for mobile display): Increase color saturation to make the firework colors more vivid and visually striking.

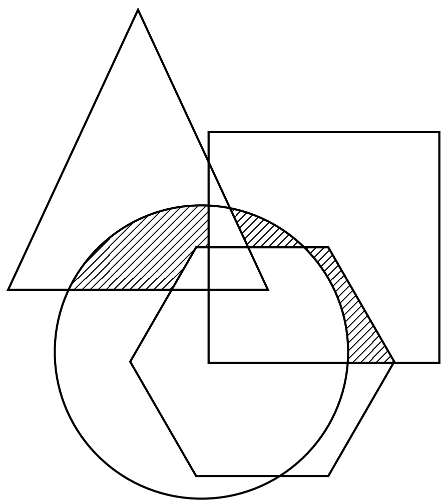
c) **Sharpening** (especially for desktop/print): Apply unsharp masking or high-boost filtering to enhance the crispness of firework spark edges.

Operations to avoid: Gamma correction ($\gamma < 1$), log transforms, and histogram equalization—these lift the dark sky and reveal noise, degrading the aesthetic quality.

Problem 2) (20 points)

A, B, C and D are four sets. Give expressions for the set shown shaded in Figure 2 in terms of A, B, C, and D. Note that all of the shaded areas constitute of one set





Step 1: Identify the Shaded Regions

Examining the diagram, the shaded area consists of three distinct regions where exactly two shapes overlap, excluding the areas where three or more shapes would overlap:

a) The region where A (triangle) and D (circle) overlap, but not B or C
b) The region where B (rectangle) and D (circle) overlap, but not A or C
c) The region where B (rectangle) and C (hexagon) overlap, but not A or D

Step 2: Write Set Expressions for Each Region

Using set intersection ($\cap$) and complement notation (X^c):

Region 1: Points in both A and D, but not in B or C:

$$A \cap D \cap B^c \cap C^c$$

Region 2: Points in both B and D, but not in A or C:

$$B \cap D \cap A^c \cap C^c$$

Region 3: Points in both B and C, but not in A or D:

$$B \cap C \cap A^c \cap D^c$$

Step 3: Combine Using Union

The complete shaded region is the union of all three regions.

Results

The shaded set can be expressed as:

$$\boxed{(A \cap D \cap B^c \cap C^c) \cup (B \cap D \cap A^c \cap C^c) \cup (B \cap C \cap A^c \cap D^c)}$$

Problem 3) (25 points)

Given a 3x3 spatial filter $\frac{1}{9}\begin{bmatrix} -1 & -1 & -1 \\ -1 & -17 & -1 \\ -1 & -1 & -1 \end{bmatrix}$ answer the following questions:

a) (10 points) What type of filter is it? When applying this filter on an image, what kind of effect you will expect in the output image? Why? (Hint: is this filter an image smoothing filter, an image sharpening filter or an edge detector?)

b) (15 points) Perform the image convolution using this filter on a 3x3 image $\begin{bmatrix} 3 & 5 & 12 \\ 7 & 18 & 11 \\ 5 & 9 & 15 \end{bmatrix}$.

Give the CROPPED filtering result for the output image. (Hint: you need to assume an appropriate zero mapping)

Step 1: Analyze the Filter Type

The filter $\frac{1}{9}\begin{bmatrix} -1 & -1 & -1 \\ -1 & -17 & -1 \\ -1 & -1 & -1 \end{bmatrix}$ can be decomposed as:

$$= -\frac{1}{9}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} - \frac{16}{9}\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} = -(\text{averaging}) - \frac{16}{9}(\text{identity})$$

This is a **sharpening filter** (high-boost filter) with inversion:

- The center weight magnitude (17) is much larger than neighbor weights (1), emphasizing the center pixel and enhancing edges
- All negative coefficients mean the output is inverted (negative image)
- Sum of coefficients $= \frac{1}{9}(-8 - 17) = -\frac{25}{9}$, so constant regions are scaled by $-\frac{25}{9} \approx -2.78$

Step 2: Apply Zero Padding

Pad the image with zeros around the border:

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 3 & 5 & 12 & 0 \\ 0 & 7 & 18 & 11 & 0 \\ 0 & 5 & 9 & 15 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Step 3: Compute Convolution — Row 0

For each output pixel, multiply the filter by the corresponding 3×3 neighborhood and sum. The filter coefficients are -1 for all neighbors and -17 for the center, with a normalization factor of $\frac{1}{9}$.

Position (0,0): Neighborhood $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 3 & 5 \\ 0 & 7 & 18 \end{bmatrix}$

$$\frac{1}{9}[(-1)(0+0+0+0+5+0+7+18)+(-17)(3)] = \frac{1}{9}[-30-51] = \frac{-81}{9} = -9$$

Position (0,1): Neighborhood $\begin{bmatrix} 0 & 0 & 0 \\ 3 & 5 & 12 \\ 7 & 18 & 11 \end{bmatrix}$

$$\frac{1}{9}[(-1)(0+0+0+3+12+7+18+11)+(-17)(5)] = \frac{1}{9}[-51-85] = \frac{-136}{9} \approx -15.1$$

Position (0,2): Neighborhood $\begin{bmatrix} 0 & 0 & 0 \\ 5 & 12 & 0 \\ 18 & 11 & 0 \end{bmatrix}$

$$\frac{1}{9}[(-1)(0+0+0+5+0+18+11+0)+(-17)(12)] = \frac{1}{9}[-34-204] = \frac{-238}{9} \approx -26.4$$

Step 4: Compute Convolution — Row 1

Position (1,0): Neighborhood $\begin{bmatrix} 0 & 3 & 5 \\ 0 & 7 & 18 \\ 0 & 5 & 9 \end{bmatrix}$

$$\frac{1}{9}[(-1)(0+3+5+0+18+0+5+9)+(-17)(7)] = \frac{1}{9}[-40-119] = \frac{-159}{9} \approx -17.7$$

Position (1,1): Neighborhood $\begin{bmatrix} 3 & 5 & 12 \\ 7 & 18 & 11 \\ 5 & 9 & 15 \end{bmatrix}$

$$\frac{1}{9}[(-1)(3+5+12+7+11+5+9+15)+(-17)(18)] = \frac{1}{9}[-67-306] = \frac{-373}{9} \approx -41.4$$

Position (1,2): Neighborhood $\begin{bmatrix} 5 & 12 & 0 \\ 18 & 11 & 0 \\ 9 & 15 & 0 \end{bmatrix}$

$$\frac{1}{9}[(-1)(5+12+0+18+0+9+15+0)+(-17)(11)] = \frac{1}{9}[-59-187] = \frac{-246}{9} \approx -27.3$$

Step 5: Compute Convolution — Row 2

Position (2,0): Neighborhood $\begin{bmatrix} 0 & 7 & 18 \\ 0 & 5 & 9 \\ 0 & 0 & 0 \end{bmatrix}$

$$\frac{1}{9}[(-1)(0+7+18+0+9+0+0+0)+(-17)(5)] = \frac{1}{9}[-34-85] = \frac{-119}{9} \approx -13.2$$

Position (2,1): Neighborhood $\begin{bmatrix} 7 & 18 & 11 \\ 5 & 9 & 15 \\ 0 & 0 & 0 \end{bmatrix}$

$$\frac{1}{9}[(-1)(7+18+11+5+15+0+0+0)+(-17)(9)] = \frac{1}{9}[-56-153] = \frac{-209}{9} \approx -23.2$$

Position (2,2): Neighborhood $\begin{bmatrix} 18 & 11 & 0 \\ 9 & 15 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

$$\frac{1}{9}[(-1)(18+11+0+9+0+0+0+0)+(-17)(15)] = \frac{1}{9}[-38-255] = \frac{-293}{9} \approx -32.6$$

Results

Part a): This is a **sharpening filter** (high-boost filter) with inversion. It enhances edges by heavily weighting the center pixel relative to its neighbors, but produces a negative (inverted) output due to all negative coefficients.

Part b): The cropped filtering result with zero padding is:

$$\boxed{\frac{1}{9}\begin{bmatrix} -81 & -136 & -238 \\ -159 & -373 & -246 \\ -119 & -209 & -293 \end{bmatrix} = \begin{bmatrix} -9 & -15.1 & -26.4 \\ -17.7 & -41.4 & -27.3 \\ -13.2 & -23.2 & -32.6 \end{bmatrix}}$$

The negative values confirm the filter produces an inverted output.

Problem 4) (25 points)

An image with intensities in the range $r \in [0, L-1]$ has the pdf

$$p_r(r) = \begin{cases} \dfrac{4r}{3(L-1)^2} & 0 \leq r \leq L-1 \\ 0 & \text{otherwise} \end{cases}$$

where $L-1$ is the maximum intensity value. It is desired to transform the intensity levels of this image so that they will have the specified pdf

$$p_z(z) = \begin{cases} \dfrac{2z}{(L-1)^2} & 0 \leq z \leq L-1 \\ 0 & \text{otherwise} \end{cases}$$

Assume that the intensity value is continuous and find the transformation function $z = T(r)$ in terms of r and z that will accomplish this.

(Hints: The transformation function of histogram equalization is given as

$$s = T(r) = (L-1)\int_0^r p_r(w)\, dw$$

)

Step 1: Understand the Goal

We want to transform pixel intensities from distribution $p_r(r)$ to distribution $p_z(z)$. The challenge is that there is no direct formula to go from one arbitrary distribution to another.

Key insight: Histogram equalization maps *any* distribution to a uniform distribution. We can use this as a "bridge" between the two distributions.

$$\begin{tikzcd}
\text{Input } r \arrow[r, "s = T_r(r)"] \arrow[rr, bend right, dashed, "z = T(r) = T_z^{-1}(T_r(r))"'] & \text{Uniform } s \arrow[r, "z = T_z^{-1}(s)"] & \text{Output } z
\end{tikzcd}$$

Since both r and z map to the *same* uniform distribution s, the values of s must be equal. This lets us solve for z in terms of r.

Step 2: Equalize the Input Distribution $p_r(r)$

Apply histogram equalization to the input. The CDF (cumulative distribution function) gives us s:

$$\begin{aligned} s = T_r(r) &= (L-1)\int_0^r p_r(w)\,dw \\ &= (L-1)\int_0^r \frac{4w}{3(L-1)^2}\,dw \\ &= (L-1)\cdot\frac{4}{3(L-1)^2}\cdot\frac{w^2}{2}\Big|_0^r \\ &= (L-1)\cdot\frac{4}{3(L-1)^2}\cdot\frac{r^2}{2} = \boxed{\frac{2r^2}{3(L-1)}} \end{aligned}$$

Step 3: Equalize the Desired Distribution $p_z(z)$

Apply the same process to the desired output distribution:

$$\begin{aligned} s = T_z(z) &= (L-1)\int_0^z p_z(w)\,dw \\ &= (L-1)\int_0^z \frac{2w}{(L-1)^2}\,dw \\ &= (L-1)\cdot\frac{2}{(L-1)^2}\cdot\frac{w^2}{2}\Big|_0^z \\ &= (L-1)\cdot\frac{2}{(L-1)^2}\cdot\frac{z^2}{2} = \boxed{\frac{z^2}{L-1}} \end{aligned}$$

Step 4: Set Equal and Solve for z

Both expressions equal the same uniform value s. Setting them equal:

$$\underbrace{\frac{2r^2}{3(L-1)}}_{T_r(r)} = \underbrace{\frac{z^2}{L-1}}_{T_z(z)}$$

Multiply both sides by $(L-1)$:

$$\frac{2r^2}{3} = z^2$$

Take the square root (using positive root since intensities are non-negative):

$$z = \sqrt{\frac{2}{3}}\,r = \frac{\sqrt{6}}{3}\,r$$

Results

The transformation function is:

$$\boxed{z = T(r) = \sqrt{\frac{2}{3}}\,r = \frac{\sqrt{6}}{3}\,r \approx 0.816\,r}$$

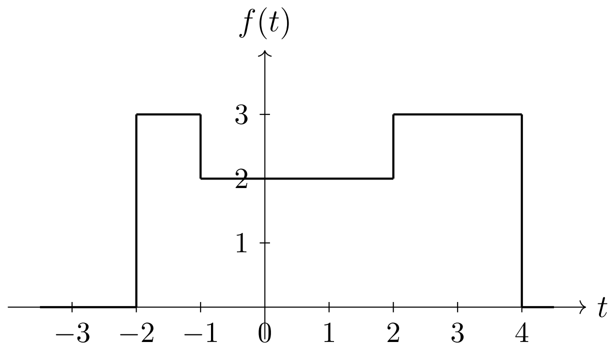
Interpretation: This is a simple linear scaling that compresses intensities by about 18%. Every input intensity r maps to a slightly smaller output intensity $z \approx 0.816r$.

Why does this work? The input PDF $p_r(r) = \frac{4r}{3(L-1)^2}$ rises more steeply than the desired PDF $p_z(z) = \frac{2z}{(L-1)^2}$. To "spread out" the distribution, we compress the intensity values, which reduces the slope of the resulting histogram.

Bonus Question (20 points)

Perform the Fourier transform to a function as shown in Figure 3.

(Hint: the definition of the Fourier transform of a function is $F(u) = \int_{-\infty}^{\infty} f(t)e^{-j2\pi ut}\,dt$)



Step 1: Express the Function Piecewise

From the figure, the function $f(t)$ is:

$$f(t) = \begin{cases} 3 & -2 \le t < -1 \\ 2 & -1 \le t < 2 \\ 3 & 2 \le t < 4 \\ 0 & \text{otherwise} \end{cases}$$

Step 2: Key Formula for Rectangular Pulse

For a constant h over interval $[a, b]$, the Fourier transform is:

$$\boxed{\int_a^b h\,e^{-j2\pi ut}\,dt = \frac{h}{j2\pi u}\left(e^{-j2\pi ua} - e^{-j2\pi ub}\right)}$$

Cheat sheet version: Just plug in height h, left bound a, right bound b. No integration needed.

Step 3: Apply the Formula to Each Segment

Segment 1: $h = 3$, $a = -2$, $b = -1$

$$F_1(u) = \frac{3}{j2\pi u}\left(e^{-j2\pi u(-2)} - e^{-j2\pi u(-1)}\right) = \frac{3}{j2\pi u}\left(e^{j4\pi u} - e^{j2\pi u}\right)$$

Segment 2: $h = 2$, $a = -1$, $b = 2$

$$F_2(u) = \frac{2}{j2\pi u}\left(e^{-j2\pi u(-1)} - e^{-j2\pi u(2)}\right) = \frac{2}{j2\pi u}\left(e^{j2\pi u} - e^{-j4\pi u}\right)$$

Segment 3: $h = 3$, $a = 2$, $b = 4$

$$F_3(u) = \frac{3}{j2\pi u}\left(e^{-j2\pi u(2)} - e^{-j2\pi u(4)}\right) = \frac{3}{j2\pi u}\left(e^{-j4\pi u} - e^{-j8\pi u}\right)$$

Step 4: Combine and Simplify

Sum all three contributions and factor out $\frac{1}{j2\pi u}$:

$$\begin{aligned} F(u) &= \frac{1}{j2\pi u}\left[3e^{j4\pi u} - 3e^{j2\pi u} + 2e^{j2\pi u} - 2e^{-j4\pi u} + 3e^{-j4\pi u} - 3e^{-j8\pi u}\right] \\ &= \frac{1}{j2\pi u}\left[3e^{j4\pi u} + (-3+2)e^{j2\pi u} + (-2+3)e^{-j4\pi u} - 3e^{-j8\pi u}\right] \\ &= \frac{1}{j2\pi u}\left[3e^{j4\pi u} - e^{j2\pi u} + e^{-j4\pi u} - 3e^{-j8\pi u}\right] \end{aligned}$$

Results

The Fourier transform is:

$$\boxed{F(u) = \frac{1}{j2\pi u}\left[3e^{j4\pi u} - e^{j2\pi u} + e^{-j4\pi u} - 3e^{-j8\pi u}\right], \quad u \neq 0}$$

For $u = 0$, compute the area directly:

$$F(0) = \int_{-\infty}^{\infty} f(t)\,dt = 3(1) + 2(3) + 3(2) = 15$$